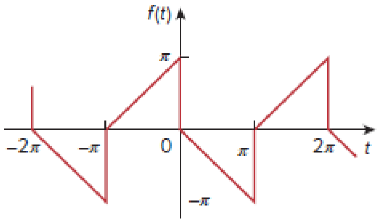
Chapter 17, Problem 29P

Problem

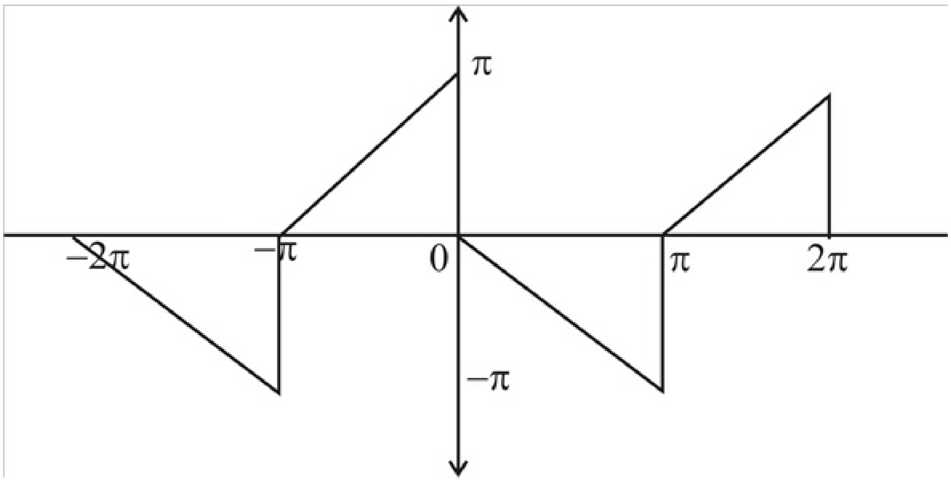
Determine the Fourier series expansion of the sawtooth function in Fig. 17.67.

Figure 17.67:



Step-by-step solution

Step 1 of 2



Step 2 of 2

$$f(t)=\begin{cases} t+\pi & -\pi<t<0 \\ -t & 0<t<\pi \end{cases}$$

$T=2\pi \qquad \omega_o=1$

The above function is half wave symmetric.

therefore $a_o=0$

$$a_n=\frac{4}{2\pi}\int_o^{\pi} f(t)\,n\omega_o t\ dt$$

$$=\frac{2}{\pi}\int_o^{\pi} -t\cos nt\,dt=\frac{-2}{\pi}\left[\frac{\sin nt}{n}\int\frac{\sin nt}{n}dt\right]$$

$$=\frac{-2}{\pi}\left[\frac{t\sin nt}{n}+\frac{\cos nt}{n^2}\right]_0^{\pi}$$

$$=\frac{2}{\pi}\left[\frac{\cos n\pi}{n^2}-\frac{1}{n^2}\right]=\frac{-2}{\pi}\left[\frac{(-1)^n-1}{n^2}\right]$$

$$\left.\begin{array}{ll} a_n=\dfrac{4}{n^2\pi} & n=odd \\ \quad 0 & n=even \end{array}\right\}$$

$$b_n=\frac{4}{T}\int_o^{\frac{T}{2}} f(t)\sin n\omega_o t\ dt=\frac{2}{\pi}\int_o^{\pi} f(t)\sin n\omega_o t\,dt$$

$$=\frac{-2}{\pi}\int_o^{\pi} t\sin nt\ dt=\frac{-2}{\pi}\left[\frac{-t\cos nt}{n}+\int\frac{\cos nt}{n}\right]$$

$$=\frac{-2}{\pi}\left[\frac{-t\cos nt}{n}+\frac{\sin nt}{n^2}\right]_o^{\pi}$$

$$=\frac{-2}{\pi}\left[\frac{-t\cos n\pi+0}{n}\right]$$

$$b_n=\frac{-2\pi}{n\pi}(-1)^2=\frac{-2}{\pi}(-1)^n$$

$$\boxed{f(t)=2\sum_{n=1}^{\infty}\left[\frac{2}{n^2\pi}\cos(nt)-\frac{1}{n}\sin(nt)\right]\ n=2k-1}$$